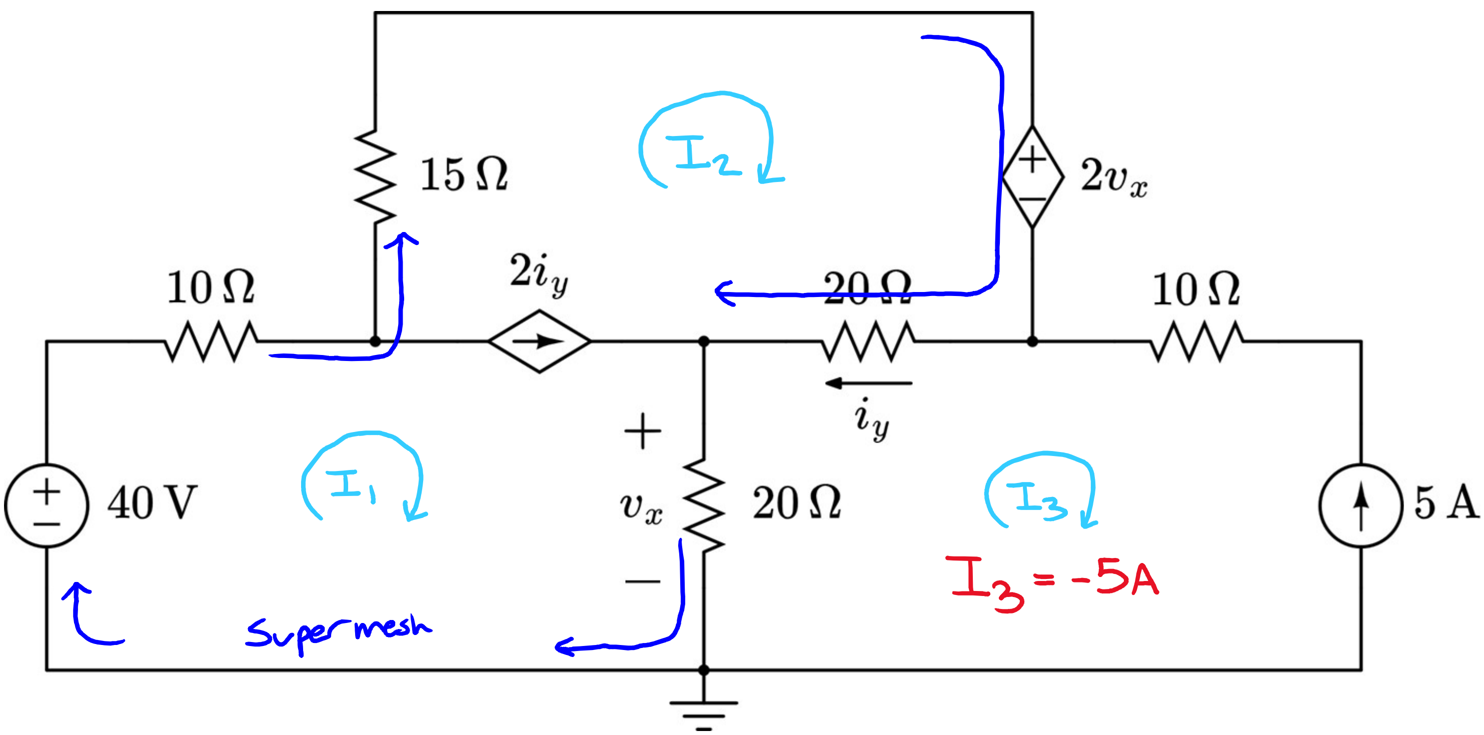
Practice Quiz Five Solution: Mesh Current Analysis (Hard)

For the circuit shown below, use mesh current analysis to determine the value of both dependent sources. Your solution must explicitly define and solve for each mesh current.

Step One: Define each mesh current, identify supermesh



Supermesh: $I_1 - I_2 = 2I_y$

Direct Substitution: $I_1 - I_2 = 2(I_2 - I_3)$

$$I_1 - I_2 = 2I_2 - 2I_3$$

$$-2I_2 + 2I_3$$

$$I_1 - 3I_2 + 2I_3 = 0$$

$$I_1 - 3I_2 = 10$$

V_x, I_y Relationship:

$$I_y = I_2 - I_3$$

$$V_x = 20(I_1 - I_3)$$

Step Two: KVL following the Supermesh:

$$-40 + 10I_1 + 15I_2 + \underline{2V_x} + 20(I_2 - I_3) + \underline{V_x} = 0$$

$$+40 \qquad\qquad +40$$

$$10I_1 + \underline{15I_2} + 3V_x + \underline{20I_2} - 20I_3 = 40$$

$$10I_1 + 35I_2 + 3(20I_1 - 20I_3) - 20I_3 = 40$$

$$10I_1 + 35I_2 + 60I_1 - 60I_3 - 20I_3 = 40$$

$$70I_1 + 35I_2 - 80I_3 = 40$$

$$I_3 = -5A$$

$$70I_1 + 35I_2 = -360$$

Step Three: Formulate 2x2 Matrix

$$\begin{bmatrix} 1 & -3 \\ 70 & 35 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 10 \\ -360 \end{bmatrix} \qquad \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1 & -3 \\ 70 & 35 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 10 \\ -360 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} -\frac{146}{49} \\ -\frac{212}{49} \end{bmatrix} A = \begin{bmatrix} -2.98 \\ -4.33 \end{bmatrix} A$$

Self-Grading Rubric	
Clearly name and label each mesh	+1 point
Correct KVL equation following the supermesh	+1 points
Identify $i_3 = \pm 5$ A (depending on labeled direction)	+2 points
Correct supermesh constraint equation	+2 points
Correct definition of v_x	+1 point
Correct definition of i_y	+1 point
Correct answer for i_1	+1 point
Correct answer for i_2	+1 point